## 1.5 Taxonomic hierarchy

In biological classification, the taxonomical hierarchy includes seven major categories namely kingdom, phylum, class, order, family, genus and species and other intermediate categories such as subkingdom, grade, division, subdivision, subphylum, superclass, subclass, superorder, suborder, superfamily, subfamily and subspecies.

### Species

Species is the basic unit of classification in the taxonomic hierarchial system. It is a group of animals having similar morphological features (traits) and is reproductively isolated to produce fertile offspring. There are some exceptional animals which can produce **sterile offspring** because of mating with closely related species (Figure 1.2).

**Genus:** It is a group of closely related species which have evolved from a common ancestor. In some genus there is only one species which is called as **monotypic genus** (e.g. Red panda is the only species in the genus *Ailurus* : *Ailurus fulgens*). If there are more than one species in the genus it is known as **polytypic genus,** for example 'cats' come under the Genus *Felis*, which has a number of closely related species, *Felis domestica* (domestic cat), *Felis margarita* (jungle cat). *Felis silvestris* (wild cat)

**Family:** It is a taxonomic category which includes a group of related genera with less similarity as compared to genus and species. For example, the family Felidae includes the genus *Felis* (cats) and the genus *Panthera* (lions, tigers, leopards).

**Order:** This category includes an assemblage of one or more related families which show few common features. One or more similar families are grouped together to form an order. For example, family *Canidae* and *Felidae* are placed in the order Carnivora.

**Class**: This category includes one or more related orders with some common characters. For example order Primata comprising monkeys, apes and man is placed in the Class Mammalia, along with



Hinny



Mule



Liger



Tigon

**Figure 1.2** Sterile offsprings

**Crosses between**

Male horse and Female Donkey results in **Hinny** (Sterile).

Male Donkey and Female Horse results in **Mule** (Sterile)

Male Lion and Female Tiger results in **Liger**

Male Tiger and Female Lion results in **Tigon**



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